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A

B

C

D

E

F

G

H

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01\_carrier\_interface

File: 01\_carrier\_interface.kicad\_sch

20\_positive\_rails

File: 20\_positive\_rails.kicad\_sch

40\_power\_supervision

File: 40\_power\_supervision.kicad\_sch

04\_dac\_0\_3

File: 04\_dac\_0\_3.kicad\_sch

06\_analog\_outputs

File: 06\_analog\_outputs.kicad\_sch

10\_input\_protection

File: 10\_input\_protection.kicad\_sch

30\_negative\_reference

File: 30\_negative\_reference.kicad\_sch

03\_peripheral\_boundaries

File: 03\_peripheral\_boundaries.kicad\_sch

05\_dac\_4\_7

File: 05\_dac\_4\_7.kicad\_sch

41\_power\_status

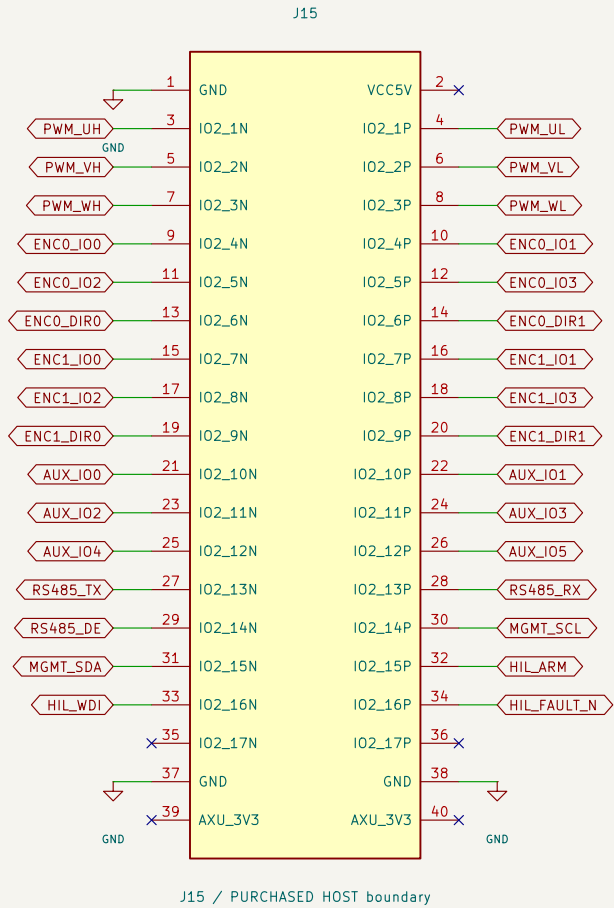
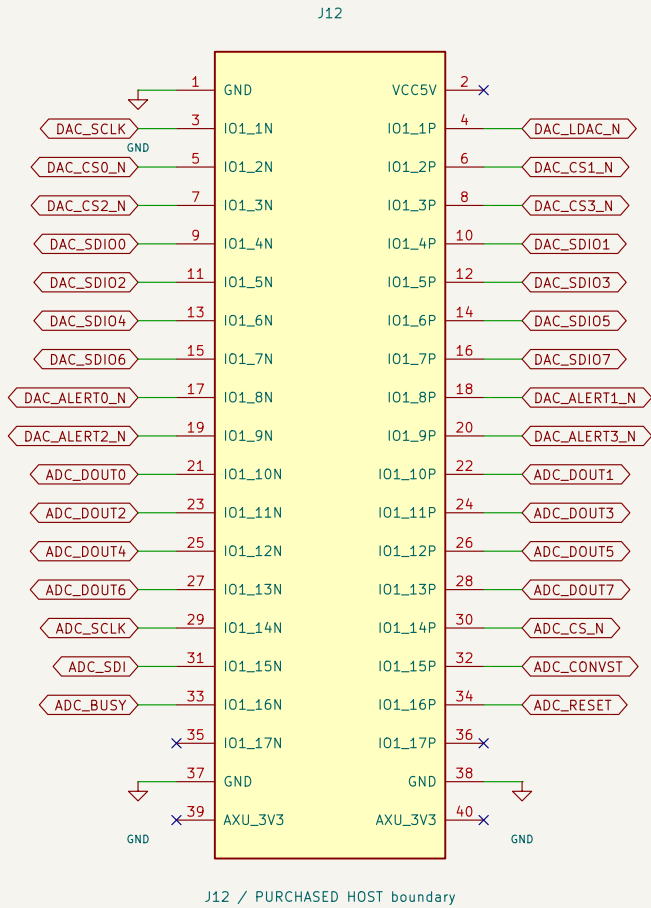
File: 41\_power\_status.kicad\_sch

Sheet: /  
File: servohl\_io\_revB.kicad\_sch

Title: ServoHIL Rev.B – native editable single–SoC expansion

Size: A2Date:KICad E.D.A. 10.0.6

Rev: B–power–developmentId: 1/11



Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB-I or -E. No board-power backfeed through contacts.

Signal-pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-att/servoHIL**

Sheet: /01\_carrier\_interface/

File: 01\_carrier\_interface.kicad\_sch

**Title: 01 carrier interface**

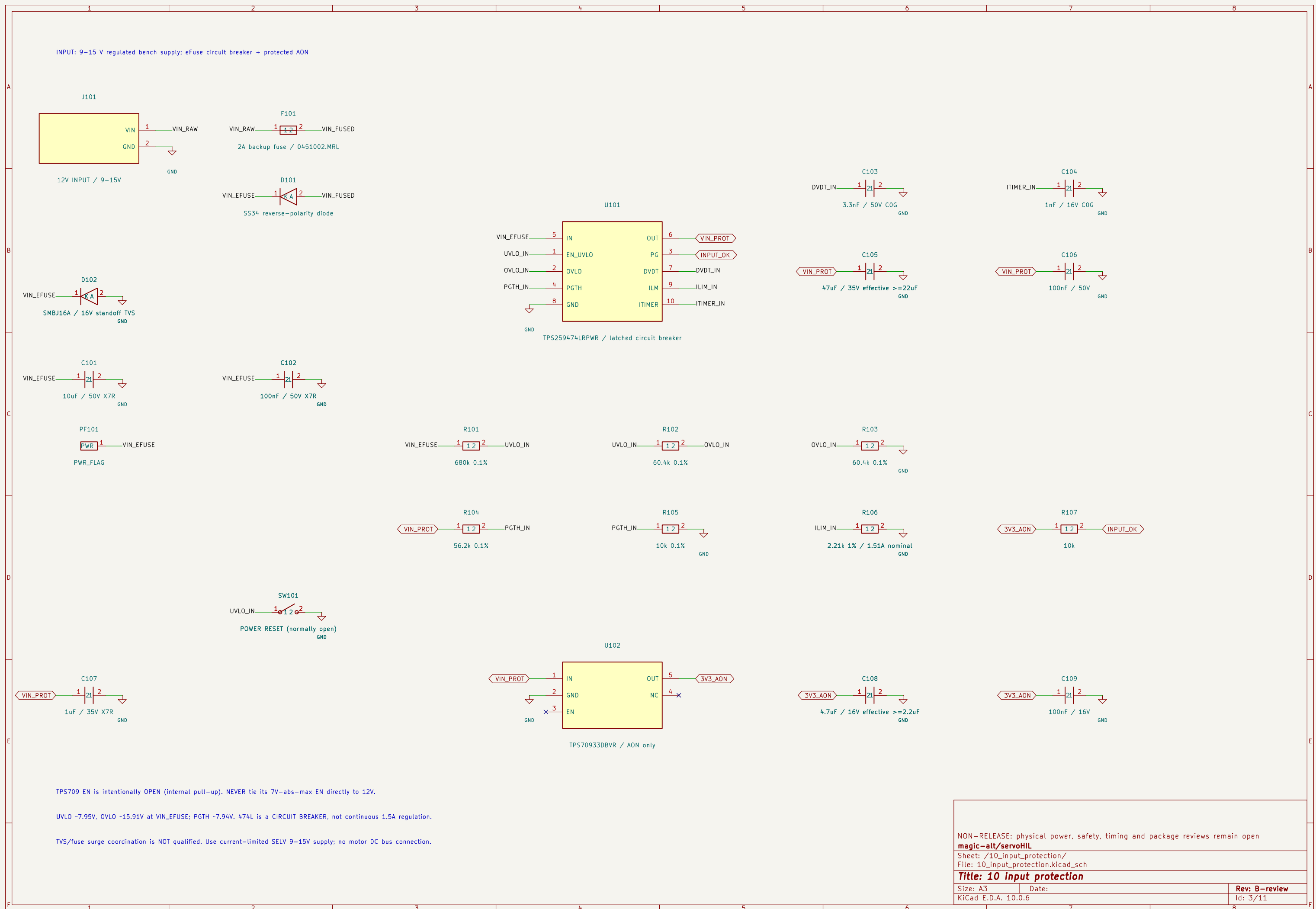
Size: A3

Date:

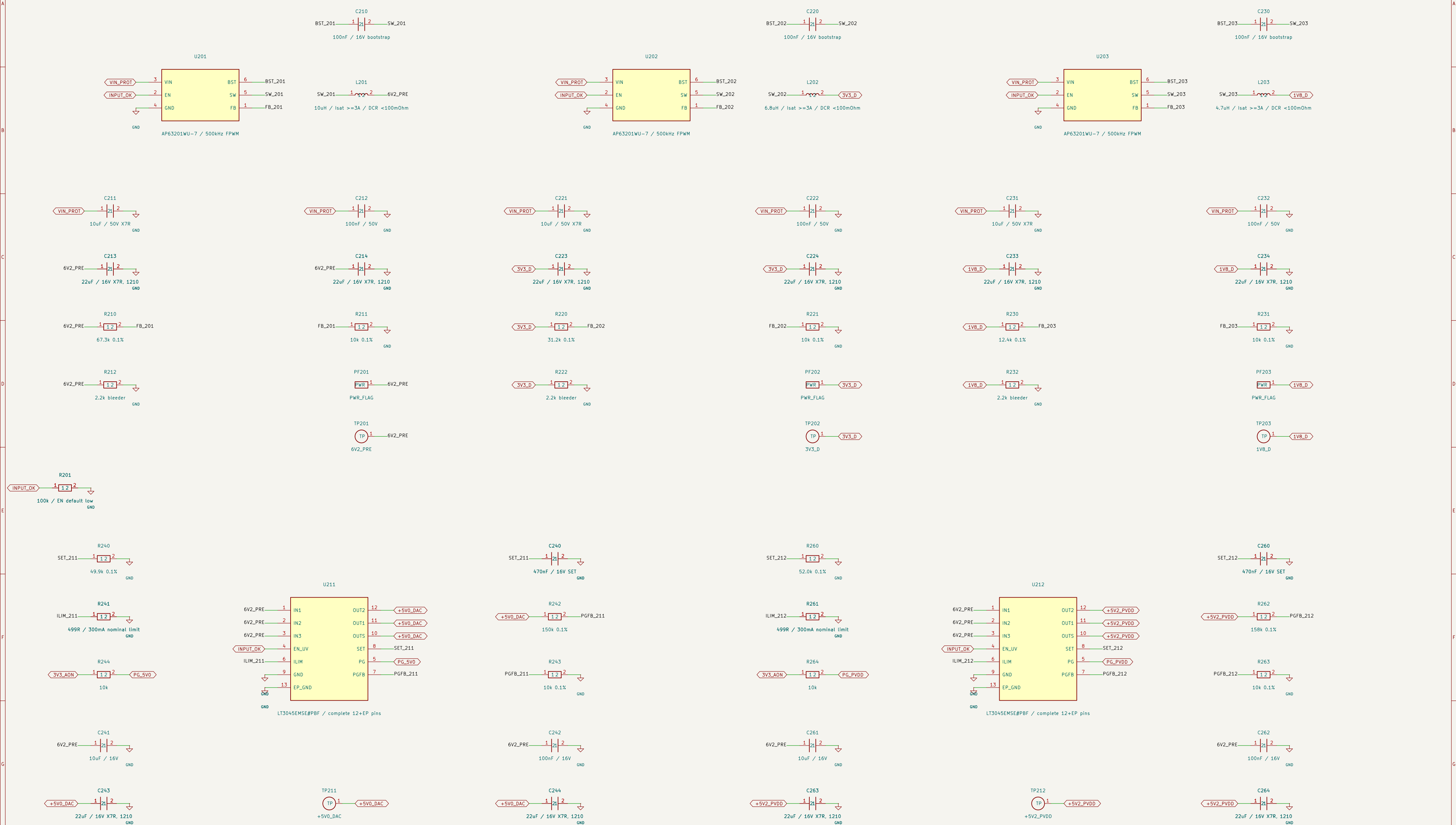
Rev: B-review

KiCad E.D.A. 10.0.6

Id: 2/11



POSITIVE POWER: 6.184V preregulator / 3.296V / 1.792V + LT3045 DAC rails



Bucks: AP63201 internal compensation; no invented COMP network. Cout requires >=20uF EFFECTIVE per buck after DC-bias/temperature.

Power budgets: 6V2\_PRE 0.50A, 3V3\_D 0.30A, 1V8\_D 0.30A; +5V0\_DAC 0.20A, +5V2\_PVDD 0.12A. NOT a 2A-per-rail board rating.

LT3045 OUTS and SET require Kelvin routing. MSE EP13=GND, +5V0 PG trip -4.80V; +5V2 PG trip -5.04V. Layout/thermal/loop response OPEN.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /20\_positive\_rails/

File: 20\_positive\_rails.kicad\_sch

**Title: 20 positive rails**

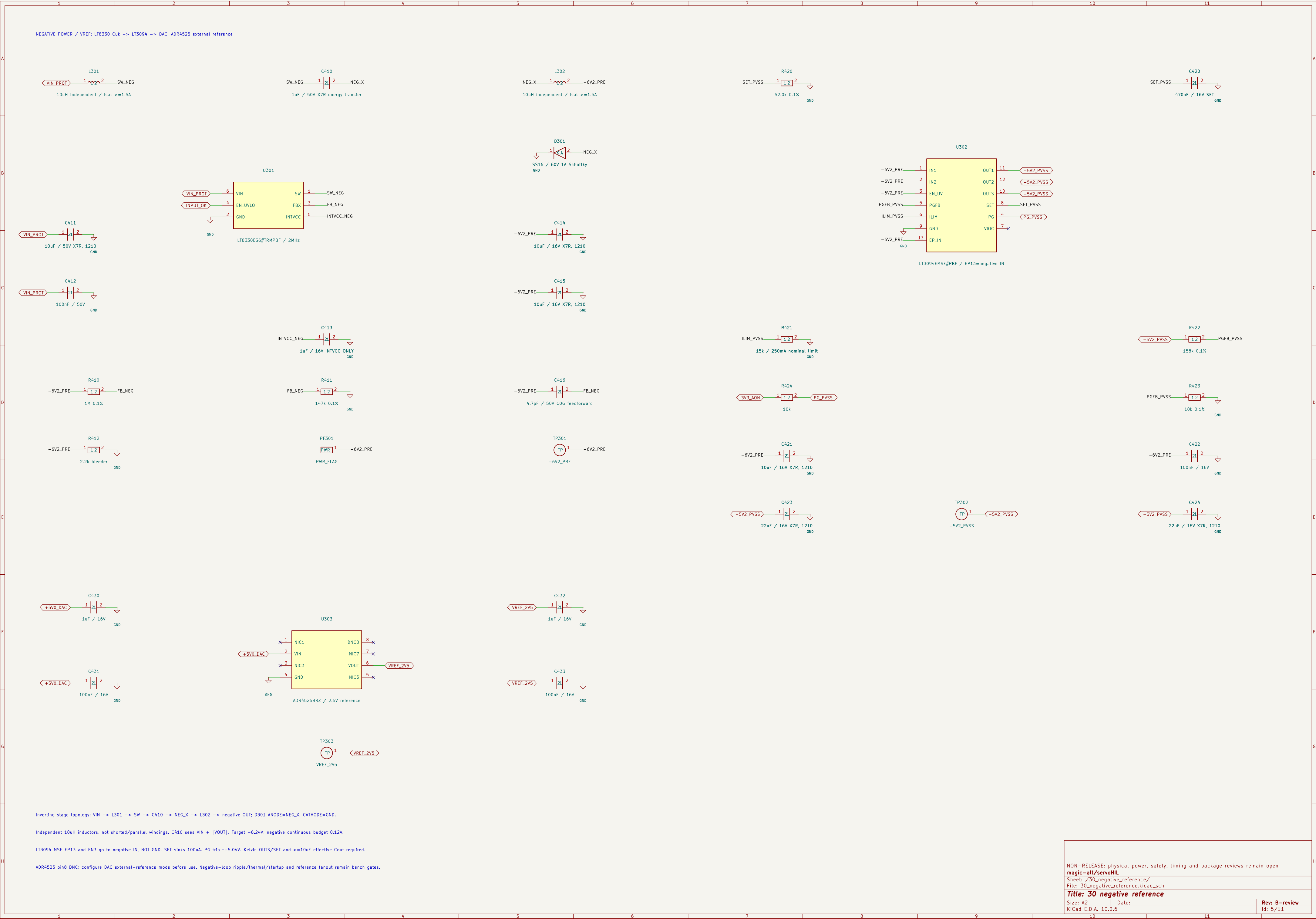
Size: A2

Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 4/11



NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /30\_negative\_reference/

File: 30\_negative\_reference.kicad\_sch

**Title: 30 negative reference**

Size: A2

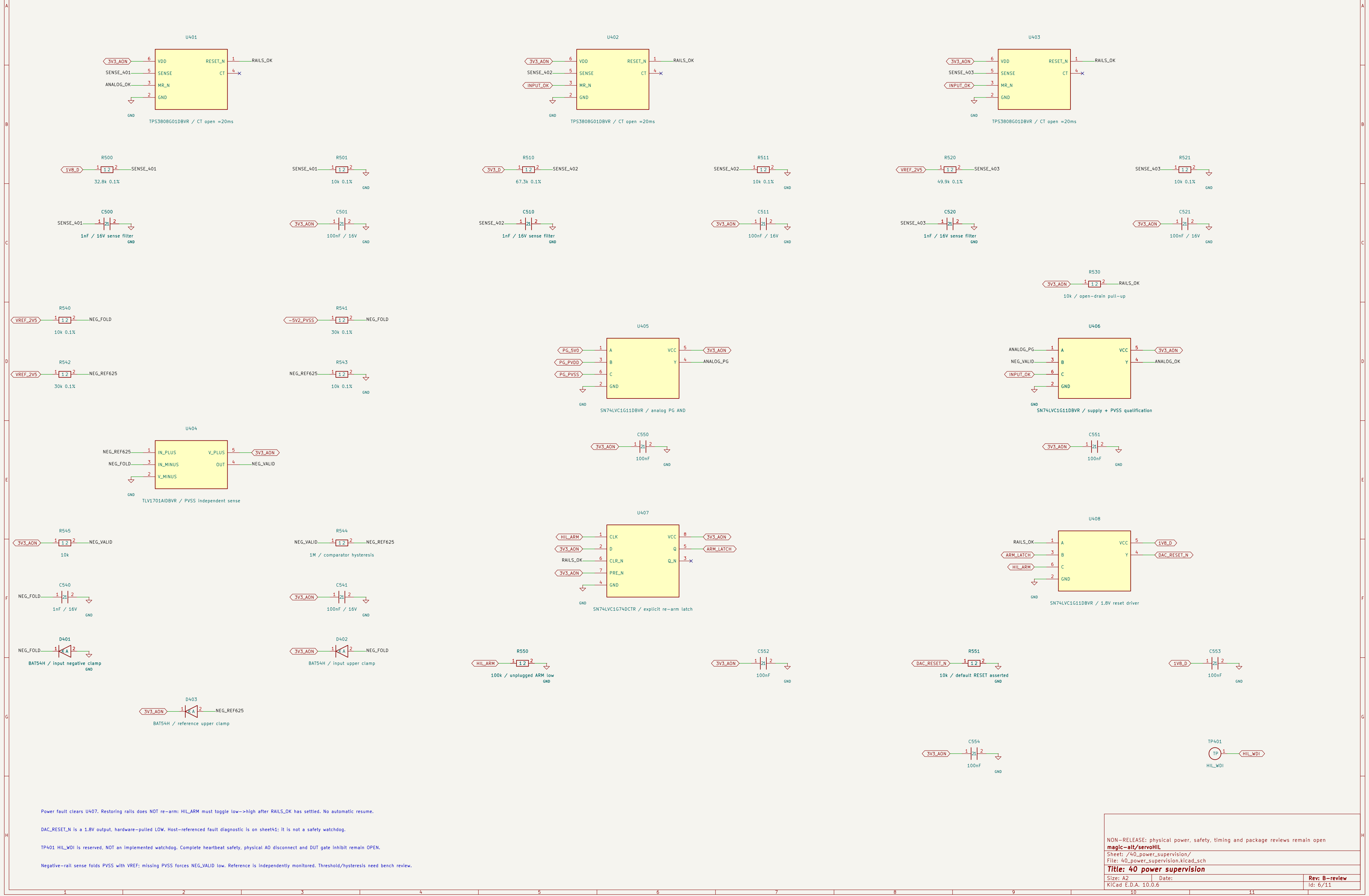
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 5/11

POWER QUALIFICATION: monitored rails, negative-rail loss detection and latched DAC reset release



Power fault clears U407. Restoring rails does NOT re-arm: HIL\_ARM must toggle low-->high after RAILS\_OK has settled. No automatic resume.

DAC\_RESET\_N is a 1.8V output, hardware-pulled LOW. Host-referenced fault diagnostic is on sheet41; it is not a safety watchdog.

TP401 HIL\_WDI is reserved, NOT an implemented watchdog. Complete heartbeat safety, physical AO disconnect and DUT gate Inhibit remain OPEN.

Negative-rail sense folds PVSS with VREF; missing PVSS forces NEG\_VALID low. Reference is independently monitored. Threshold/hysteresis need bench review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /40\_power\_supervision/

File: 40\_power\_supervision.kicad\_sch

**Title: 40 power supervision**

Size: A2

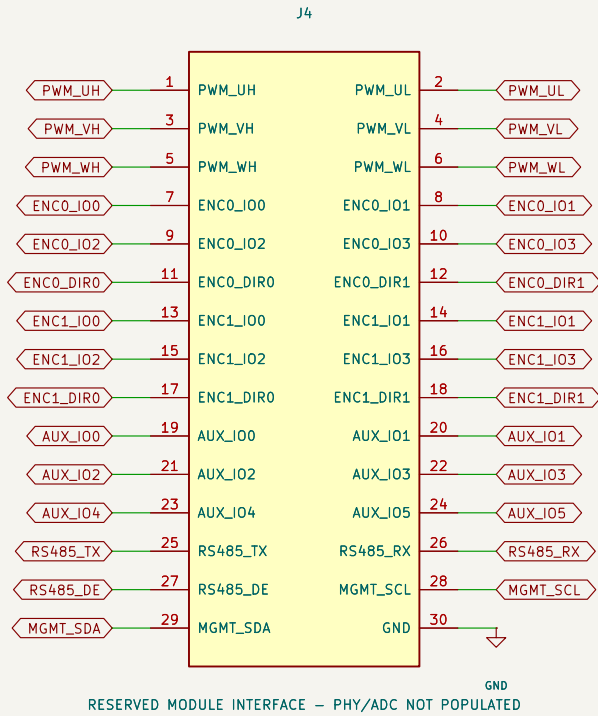
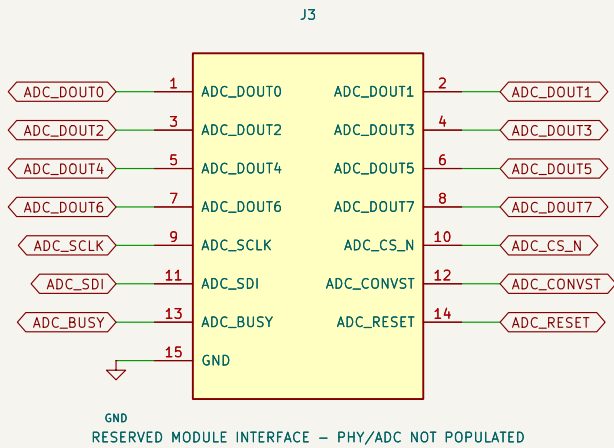
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 6/11

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

NON–RELEASE: physical power, safety, timing and package reviews remain open

**magic–alt/servoHIL**

Sheet: /03\_peripheral\_boundaries/

File: 03\_peripheral\_boundaries.kicad\_sch

**Title: 03 peripheral boundaries**

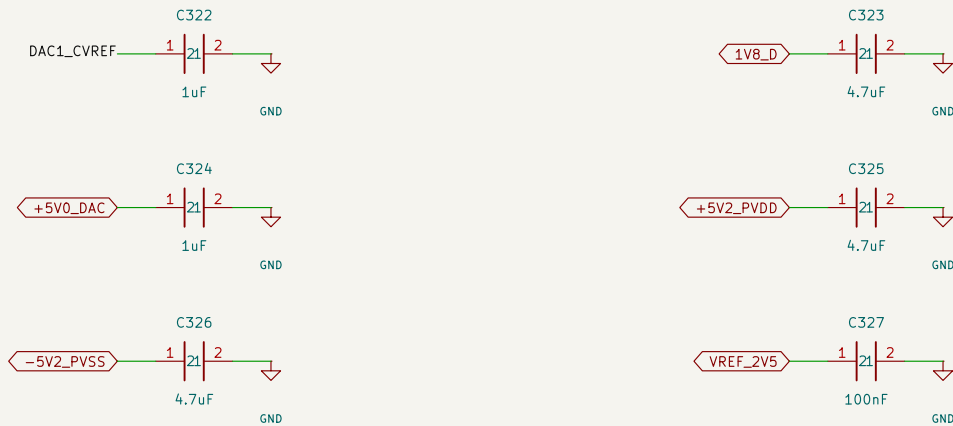
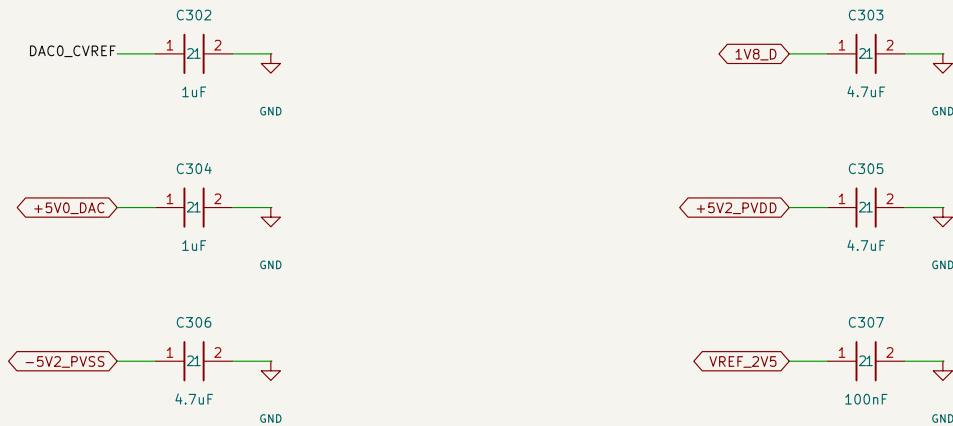
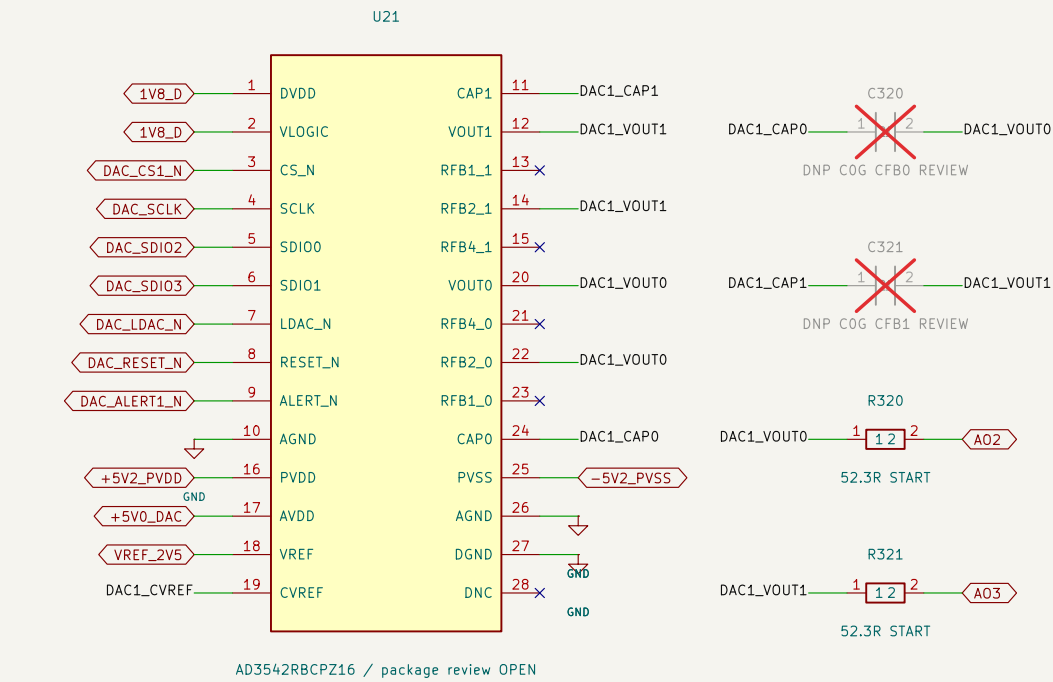
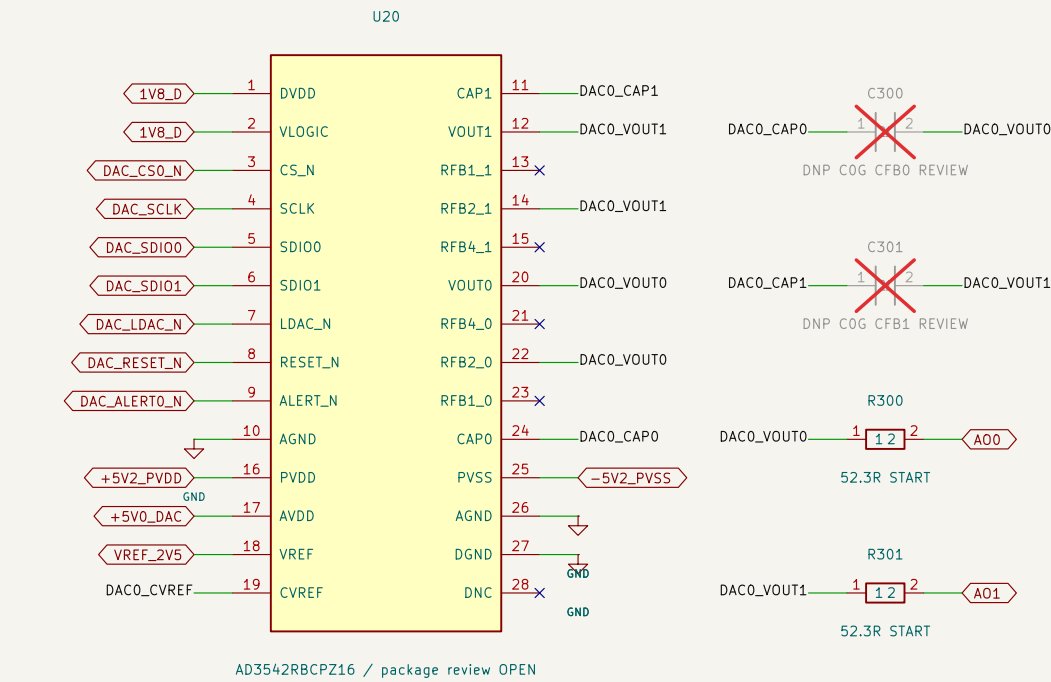
Size: A3

Date:

Rev: B–review

KiCad E.D.A. 10.0.6

Id: 7/11



RFB2 selects +/-5V capability; RFB1/RFB4/DNC are not connected, CFB and load stability are NOT approved.

External reference configuration and power–up register settings must be verified before enabling outputs.

NON–RELEASE: physical power, safety, timing and package reviews remain open

**magic–alt/servoHIL**

Sheet: /04\_dac\_0\_3/

File: 04\_dac\_0\_3.kicad\_sch

**Title: 04 dac 0 3**

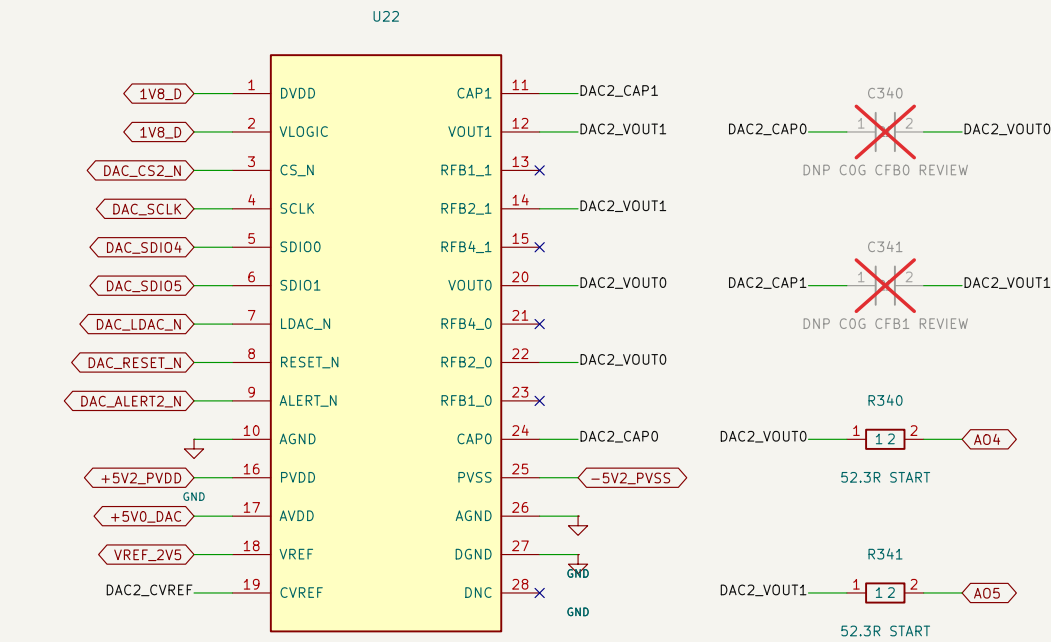
Size: A3

Date:

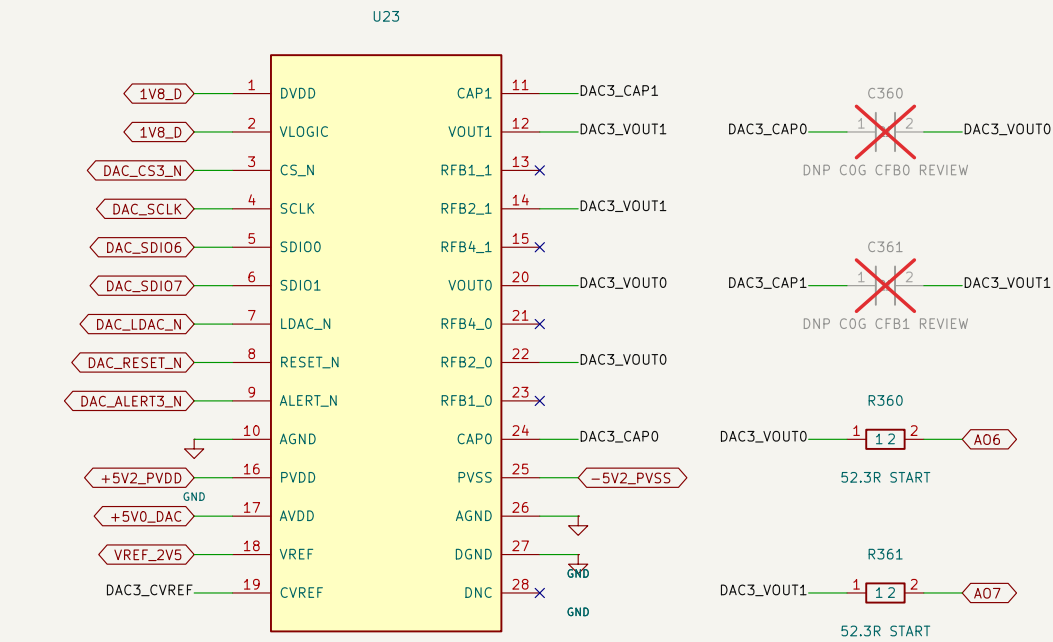
**Rev: B–review**

KiCad E.D.A. 10.0.6

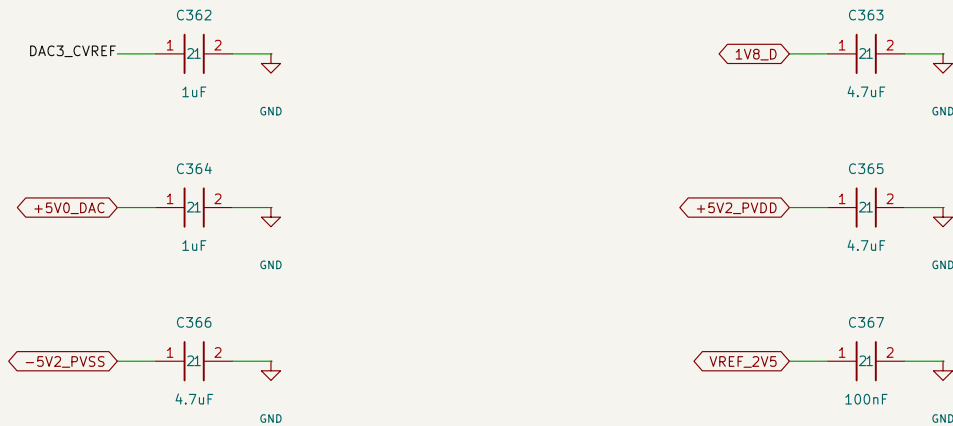
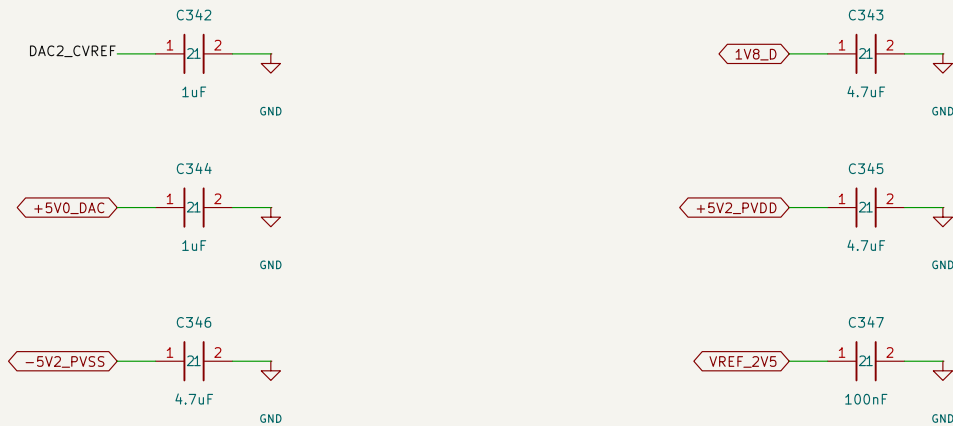
Id: 8/11



AD3542RBCPZ16 / package review OPEN



AD3542RBCPZ16 / package review OPEN



RFB2 selects +/-5V capability; RFB1/RFB4/DNC are not connected, CFB and load stability are NOT approved.

External reference configuration and power–up register settings must be verified before enabling outputs.

NON–RELEASE: physical power, safety, timing and package reviews remain open

**magic–alt/servoHIL**

Sheet: /05\_dac\_4\_7/

File: 05\_dac\_4\_7.kicad\_sch

**Title: 05 dac 4 7**

Size: A3

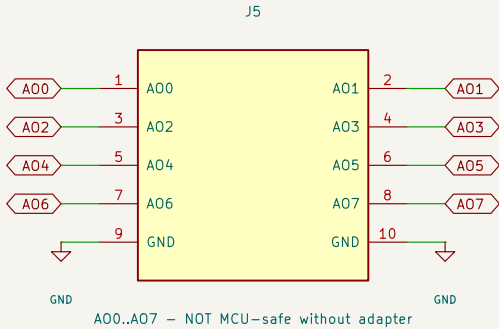
Date:

Rev: B–review

KiCad E.D.A. 10.0.6

Id: 9/11

06 – Eight analog outputs to reviewed DUT adapter



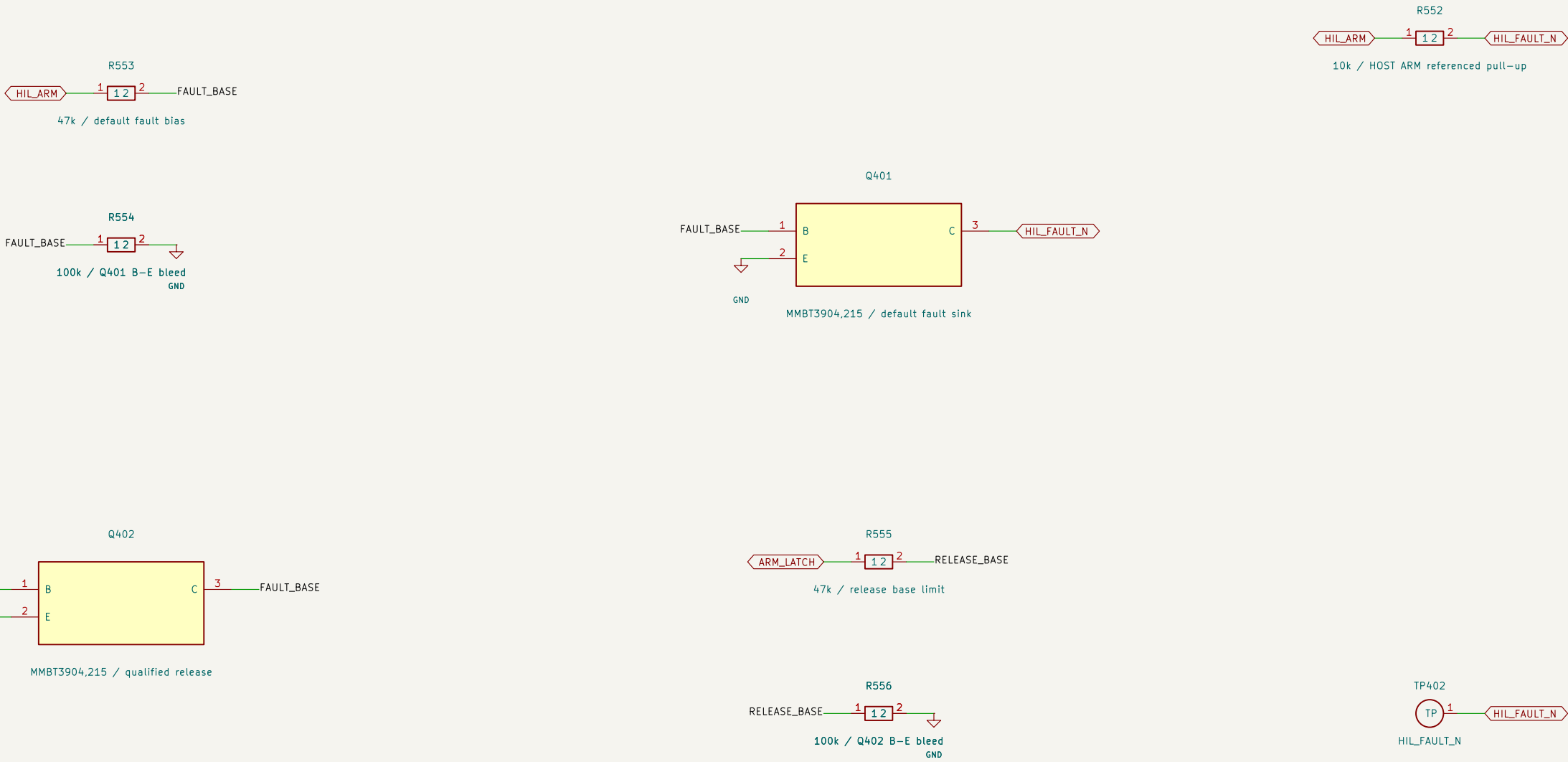
Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

NON–RELEASE: physical power, safety, timing and package reviews remain open  
**magic–alt/servoHIL**  
Sheet: /06\_analog\_outputs/  
File: 06\_analog\_outputs.kicad\_sch

|                          |       |               |
|--------------------------|-------|---------------|
| Title: 06 analog outputs |       |               |
| Size: A3                 | Date: | Rev: B–review |
| KiCad E.D.A. 10.0.6      |       | Id: 10/11     |

HOST-REFERENCED POWER STATUS: no expansion-rail pull-up on HIL\_FAULT\_N



HIL\_ARM is generated by the powered HOST, NOT an expansion rail. Q401 defaults ON while HOST requests ARM.

Only a power-qualified ARM\_LATCH turns Q402 on and releases Q401. Board power lost -> Q402 off -> Q401 reports fault.

HIL\_FAULT\_N=HIGH means power-qualified AND armed; LOW also means disarmed. At HOST off, no expansion supply drives this pin high.

Nominal fault sink <0.33mA and HOST ARM total load <0.4mA. Check transient/partial-power behavior physically; not certified safety.

Other direct DAC/ADC pins STILL require power-off backfeed protection. This diagnostic alone does not authorize hot plugging.

NON-RELEASE: physical power, safety, timing and package reviews remain open

**magic-alt/servoHIL**

Sheet: /41\_power\_status/

File: 41\_power\_status.kicad\_sch

**Title: 41 power status**

Size: A3

Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 11/11